

TEAM 002

GA ML

PIPELINE

Project Progress Report



AGENDA

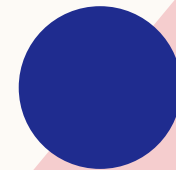
Introduction

Challenges & Roadblocks

Project Progress Report

GAN Land (alternative designs)

Finalization – How We Get There



PROJECT INTRODUCTION

- General Atomics
 - an energy and defense corporation based in San Diego, California
 - founded in 1955, acquired by the Blue family in 1986
 - specialize in unmanned aircraft, providing electro-optical, radar, signals intelligence, and automated airborne surveillance systems.
- Needs
 - GA wishes to implement data augmentation, using GANs to improve the accuracy of their machine learning implementation
 - current pipeline requires a lot of user input and manipulation
 - optimizing and increasing the machine learning model's capabilities and automation of the machine learning pipeline

PROJECT INTRODUCTION

- Objectives
 - Automating the creation and labeling of new training data.
 - Incorporating the use of generative adversarial networks for synthetic data generation and super-sampling of underrepresented training data.
 - Automating the use of training data labeling tools.
 - Automating the retraining of the AI model with new training data.
 - Automating AI model selection.
 - Automating the integration and deployment of the ideal AI models.
 - Automating AI model performance report generation.

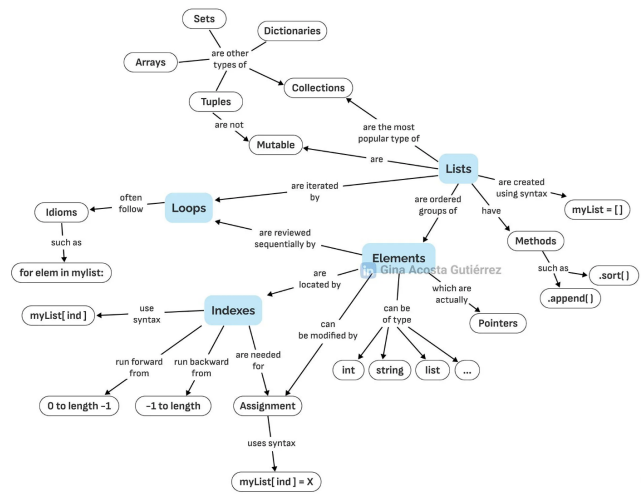
PROJECT INTRODUCTION

- Impacts
 - Individuals
 - affected in the improvement of the efficiency and speed of communications between deployed UAVs and a human commander
 - Organizations
 - reducing the cost of maintenance, labor, and the automation of their machine learning pipelines will increase the efficiency of their AI division
 - Society
 - risk of human casualties due to fatal errors from this technology

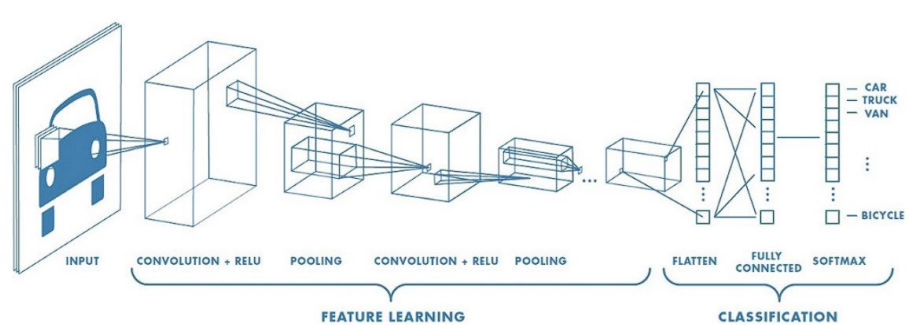
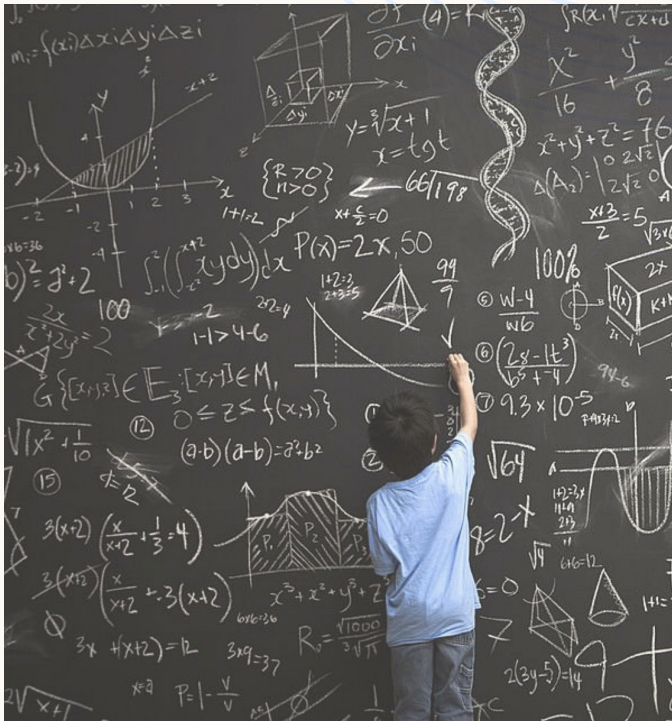
EXPERIENCE

What We Have Learned

Data Structures in Python

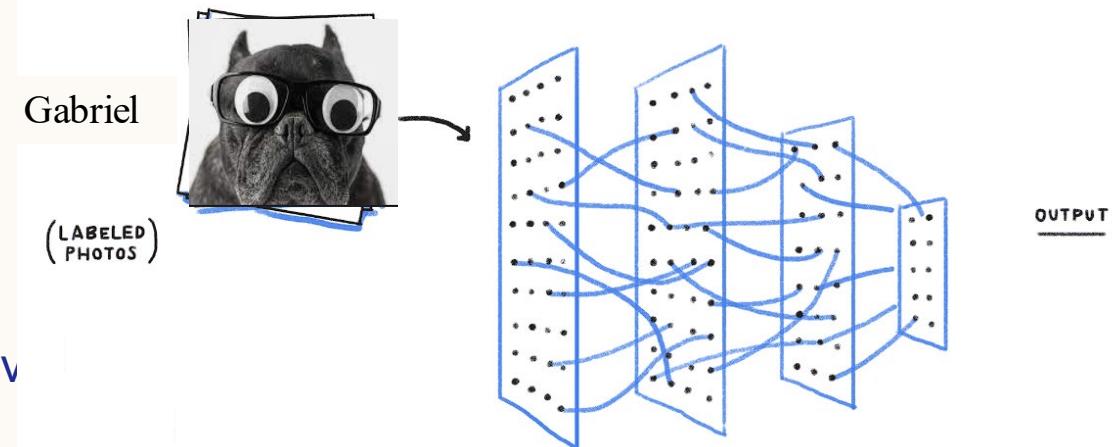


SCRIPTING
VS
PROGRAMMING



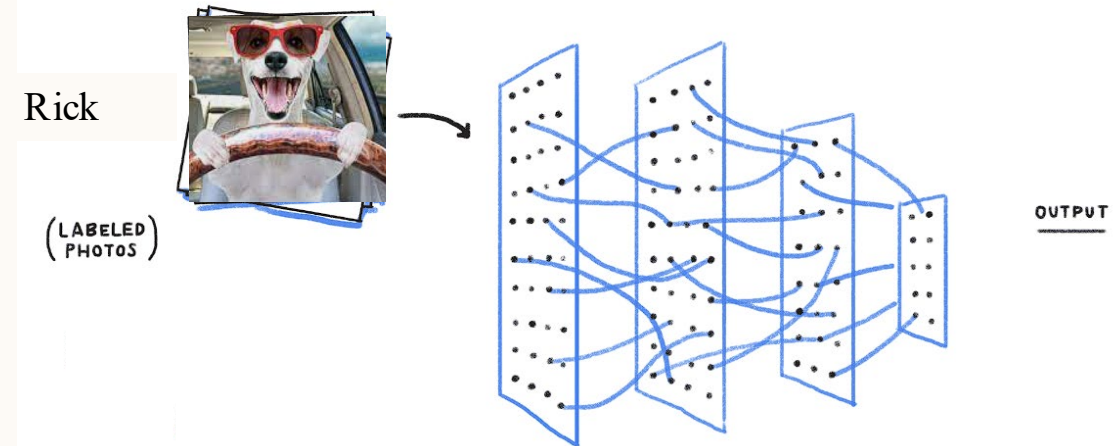
CHALLENGES AND ROADBLOCKS

- Virtual environments Anaconda (good, bad, ugly)
- Requirements and Docker
- Virtualization
- CI/ CD which requires advanced knowledge of the above
- Applying Techniques in Machine Learning & Deep Learning



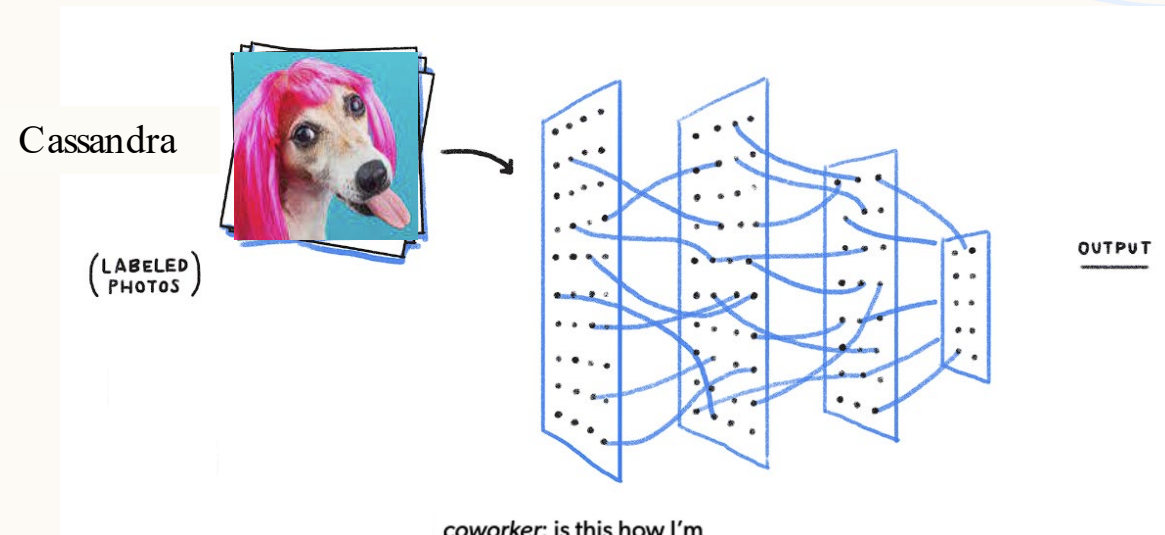
CHALLENGES AND ROADBLOCKS

- Analyzing complexities of key algorithms implemented
 - Both Training loops $O(n^3) \rightarrow 1660Ti$
 - Training and Utilities
- Summarization of lessons learned
 - Update often
 - Know when to stop
- Summarization of skills not otherwise learned
 - Deep Learning: tensor manipulation



CHALLENGES AND ROADBLOCKS

- Getting my foot in the door: How does one AI??
 - In the vast sea of AI/ML, where to start?
- Using different models together
 - Understand different models and how they can function together
- The many ways of image annotation
 - Image annotation tools, techniques



coworker: is this how I'm supposed to be doing this?

me, equally as clueless:





PRIMARY GOALS AND PROGRESS

Where have we been, where are we headed?



PROJECT PROGRESS

What We Have Done

- TensorBoard Visualization Class
- GAN model swap (several times)
- Code Base Overhaul
- Refactor, Refactor, Refactor
- Refined GAN
- Refined CNN
- User Command Line Interface
- Class Selection
- Model Evaluation Kernel Inception Distance, Frechet Inception Distance (KID, FID)
- GitLab CI Pipeline Runner
- GAN configuration
- CNN configuration

DESIGN ALTERNATIVES EXPLORED

GANland

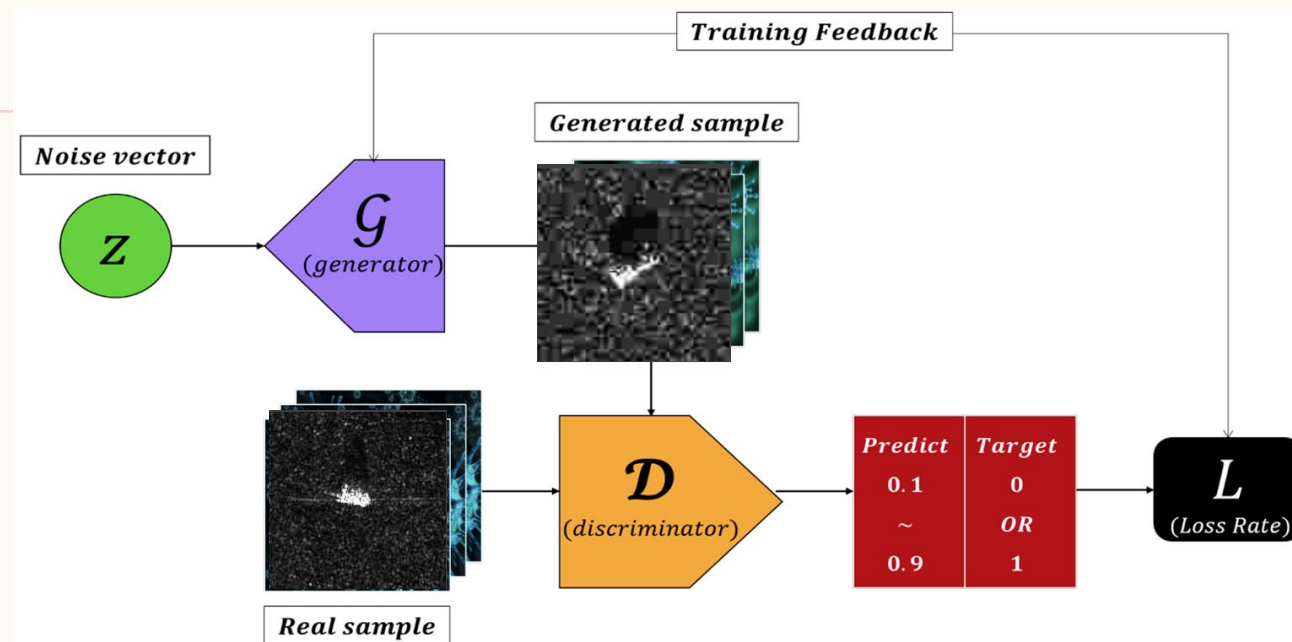
DC GAN

CGAN

AC GAN

MC GAN

MCC GAN

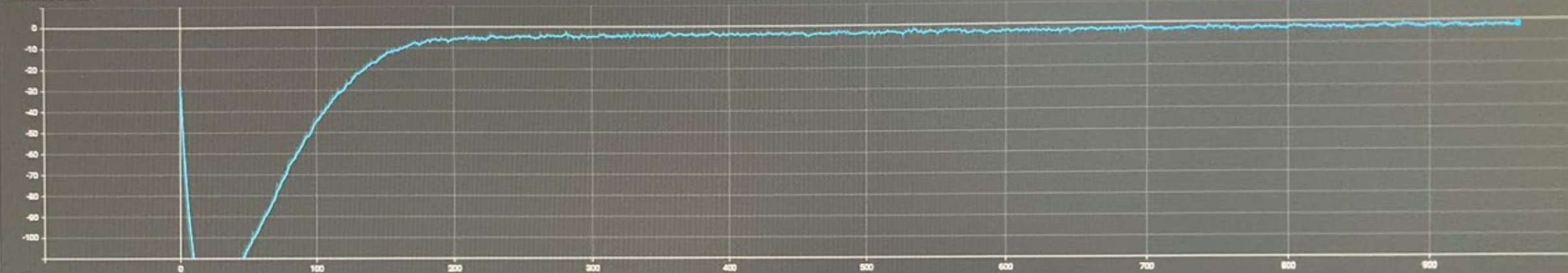


Demo Images



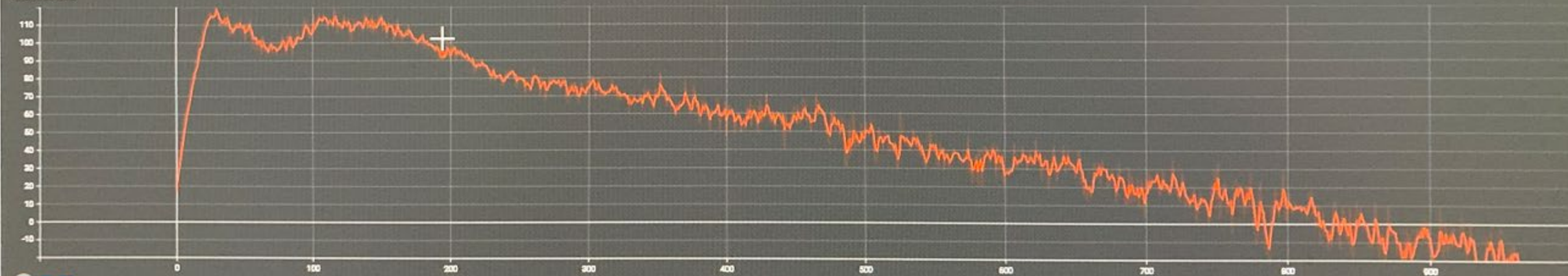
Discriminator Loss

Discriminator Loss



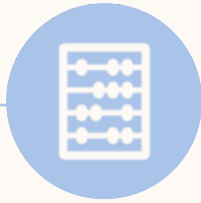
Generator Loss

Generator Loss



Name	Smoothed	Value	Step	Time	Relative
g_loss	93.41	94.23	194	Mon Apr 24, 18:27:47	8m 45s

HOW WE GET THERE



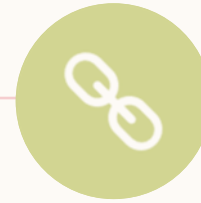
Remaining Tasks

Image Annotation
Final integration
Final system testing
CI to public space
Automation



Whos' doing what?

Cassandra
All
Rick
Gabe
All



Likelihood of completion

Low
High
High
High
Medium

SUMMARY

- What we have learned
- Skills and experiences gained
- Reflection on challenges and obstacles
- Project progress
- Design alternatives explored
- How to reach the finish line



THANK YOU

On behalf of Team 002

Cassandra Bravo

Rick Oldham

Gabriel Alfredo Siguenza

QUESTIONS?